



Maine
Department of
Education

GRADE

5

Maine Science Assessment

Released Items (2022)

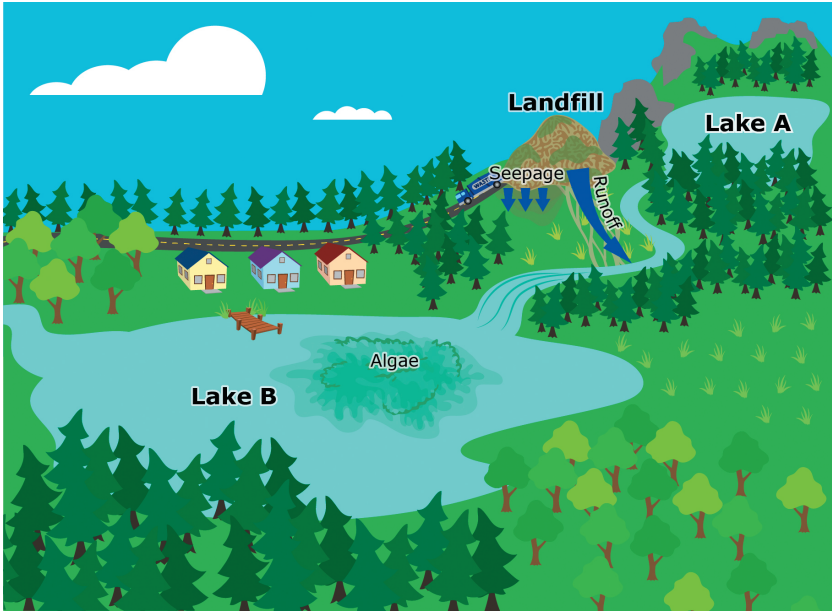


New Meridian

Use the information from Algae Bloom to answer **questions 1–5.**

Algae Bloom

Clair and Rashaun want to swim in Lake B near their homes, but it is covered with algae. However, Lake A, a few kilometers away, has clear water. Clair learns that pollutants from landfills can seep through the soil to reach the groundwater and also enter nearby streams as runoff.



Clair and Rashaun hope to find a solution to the algae problem. They listed their solutions in a table.

Solution	Image
Raised Barrier	A cross-section diagram showing a hill of 'Waste' on the right and a 'Raised Barrier' (a brick wall) on the left. The ground is brown soil.
Plastic Liner	A cross-section diagram showing a hill of 'Waste' on a 'Plastic Liner' (a blue sheet) which sits on brown soil.
Shallow Channel	A cross-section diagram showing a hill of 'Waste' on the right and a 'Shallow channel' (a V-shaped ditch) on the left, with an arrow pointing down into it.

1. Rashaun claims that the landfill is the **most likely** cause of the algae growth.

Which evidence supports this claim?

- (A) The landfill has polluted the vegetation around Lake B.
- (B) The landfill has polluted a stream that flows into Lake A.
- (C) The landfill has polluted the stream that flows into Lake B.
- (D) The landfill has polluted the groundwater that seeps into Lake A.

2. After determining how the pollution gets to Lake B, Clair wonders why pollution causes algae in the lake.

Which statement **best** describes why there are algae in Lake B?

- (A) Algae feed on the waste from human activities that pollute Lake B.
- (B) Lake A has less water and cannot hold as much pollution as Lake B.
- (C) Algae feed on the air pollution from the landfill that is blown into Lake B.
- (D) Lake A receives less pollution because it is further from the landfill than Lake B.

3. Clair and Rashaun want to model how the pollutants make their way from the landfill to the lake.

What is the correct path that the pollutants take?

Enter 1, 2, 3, or 4 into **each** box to show the order of steps in the path, with 1 being the first step and 4 the last.

Lake B

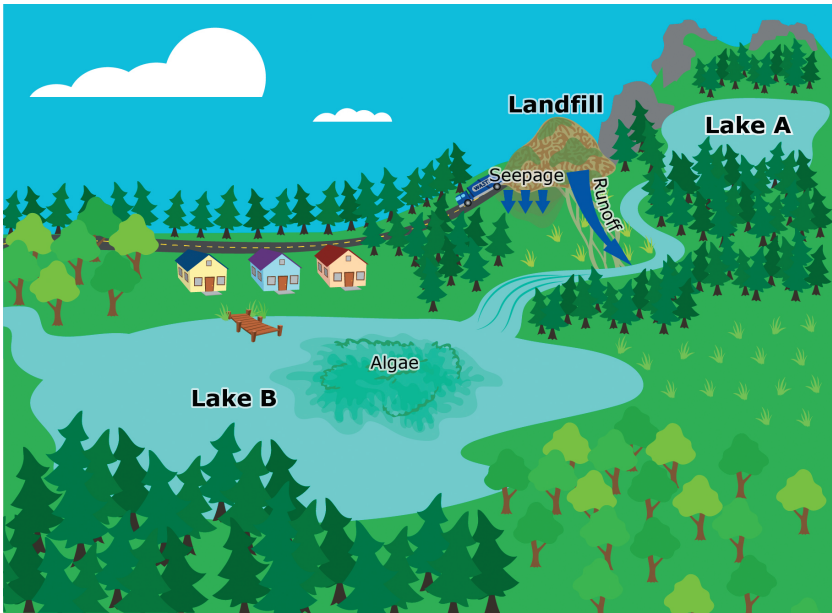
Runoff

Stream

Waste from human activities

Algae Bloom

Clair and Rashaun want to swim in Lake B near their homes, but it is covered with algae. However, Lake A, a few kilometers away, has clear water. Clair learns that pollutants from landfills can seep through the soil to reach the groundwater and also enter nearby streams as runoff.



Clair and Rashaun hope to find a solution to the algae problem. They listed their solutions in a table.

Solution	Image
Raised Barrier	A cross-section diagram showing a landfill with 'Waste' on top. A red brick wall labeled 'Raised Barrier' is built between the landfill and a body of water. The ground is brown, and the sky is blue.
Plastic Liner	A cross-section diagram showing a landfill with 'Waste' on top. A blue line labeled 'Plastic Liner' is at the base of the landfill, preventing seepage into the ground. The ground is brown, and the sky is blue.
Shallow Channel	A cross-section diagram showing a landfill with 'Waste' on top. A 'Shallow channel' is dug into the ground between the landfill and a body of water. An arrow points down into the channel. The ground is brown, and the sky is blue.

4. After considering the different solutions to the problem, Clair decides that using a raised barrier will best prevent pollutants from the landfill from reaching Lake B.

Which statement **best** states whether this solution will work?

- (A) The solution will work because a raised barrier reduces runoff.
- (B) The solution will work because a raised barrier reduces seepage.
- (C) The solution will not work because a raised barrier will not reduce runoff.
- (D) The solution will not work because a raised barrier will not reduce seepage.

- 5.** Clair and Rashaun researched different solutions to prevent the pollution in the landfill from reaching Lake B.

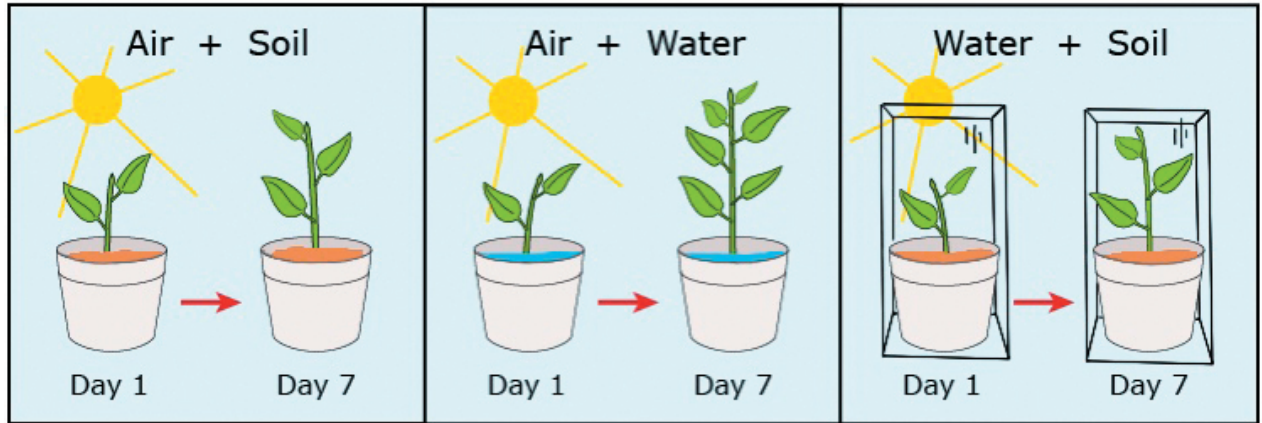
Which solution from the table is the **best** solution to prevent algae from growing in Lake B? Identify a method and explain how this would prevent algae growth in Lake B.

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6. Keaton's family has many plants inside their house and outside in their garden. Keaton sees that some plants grow faster and look healthier than others. This causes Keaton to wonder which factors affect plant growth. He decides to perform an investigation. He places three similar plants in similar pots on the same windowsill. However, he changes some of the conditions under which the plants will grow for the next seven days.

Keaton makes a poster showing the results after seven days.



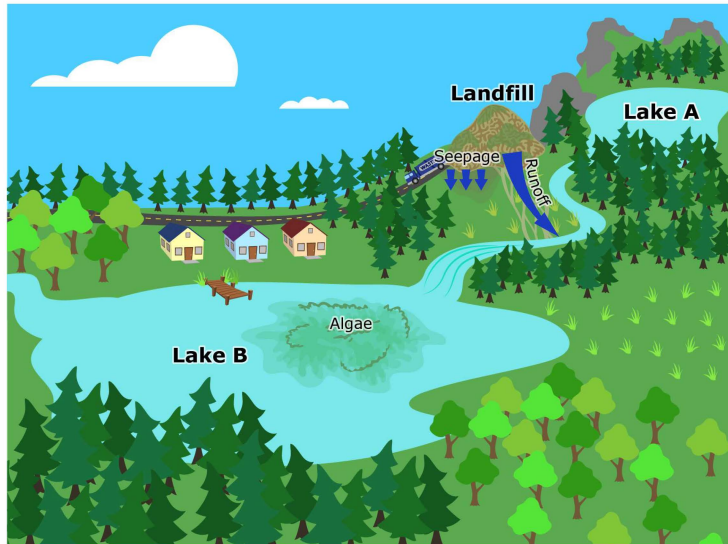
Keaton uses the data from his investigation to identify the factors that help plants grow.

Which **two** factors help plants grow the most? Choose **two**.

- (A) air
- (B) soil
- (C) sunlight
- (D) water

Algae Bloom

Clair and Rashaun want to swim in Lake B near their homes, but it is covered with algae. However, Lake A, a few kilometers away, has clear water. Clair learns that pollutants from landfills can seep through the soil to reach the groundwater and also enter nearby streams as runoff.



Clair and Rashaun hope to find a solution to the algae problem. They listed their solutions in a table.

Solution	Image
Raised Barrier	
Plastic Liner	
Shallow Channel	

1. Rashaun claims that the landfill is the most likely cause of the algae growth.

Which evidence supports this claim?

- A The landfill has polluted the vegetation around Lake B.
- B The landfill has polluted a stream that flows into Lake A.
- ☒ C The landfill has polluted the stream that flows into Lake B.
- D The landfill has polluted the groundwater that seeps into Lake A.

1 point

Standards Alignment

Discipline: Earth and Space Science

NGSS Topic: Earth's Systems

DCI: ESS3.C

Human activities in agriculture, industry, and everyday life have had major effects on the land, vegetation, streams, ocean, air, and even outer space. But individuals and communities are doing things to help protect Earth's resources and environments.

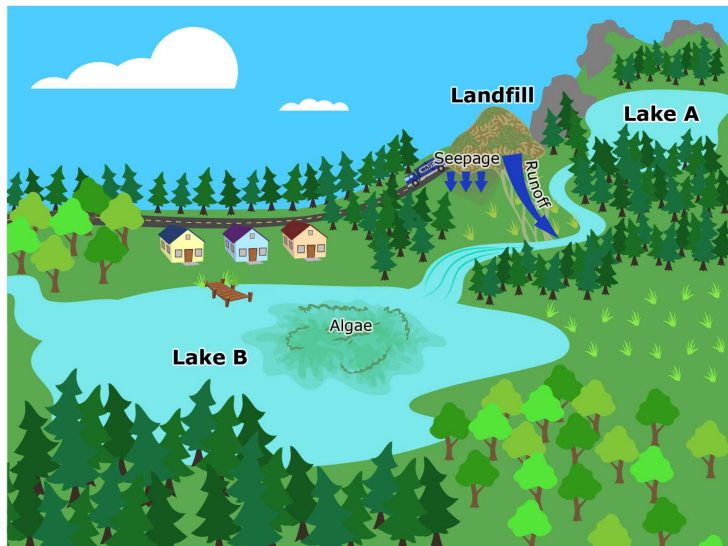
SEP2: Developing and using models

CCC4: Systems and system models

Achievement Level: Well Below State Expectations

Algae Bloom

Clair and Rashaun want to swim in Lake B near their homes, but it is covered with algae. However, Lake A, a few kilometers away, has clear water. Clair learns that pollutants from landfills can seep through the soil to reach the groundwater and also enter nearby streams as runoff.



Clair and Rashaun hope to find a solution to the algae problem. They listed their solutions in a table.

Solution	Image
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Plastic Liner	
Shallow Channel	

2. After determining how the pollution gets to Lake B, Clair wonders why pollution causes algae in the lake.

Which statement best describes why there are algae in Lake B?

- ☒ A Algae feed on the waste from human activities that pollute Lake B.
- ☐ B Lake A has less water and cannot hold as much pollution as Lake B.
- ☐ C Algae feed on the air pollution from the landfill that is blown into Lake B.
- ☐ D Lake A receives less pollution because it is further from the landfill than Lake B.

1 point

Standards Alignment

Discipline: Life Science

NGSS Topic: Matter and Energy in Organisms and Ecosystems

DCI: LS2.B

Matter cycles between the air and soil and among plants, animals, and microbes as these organisms live and die. Organisms obtain gases and water from the environment and release waste matter (gas, liquid, or solid) back into the environment.

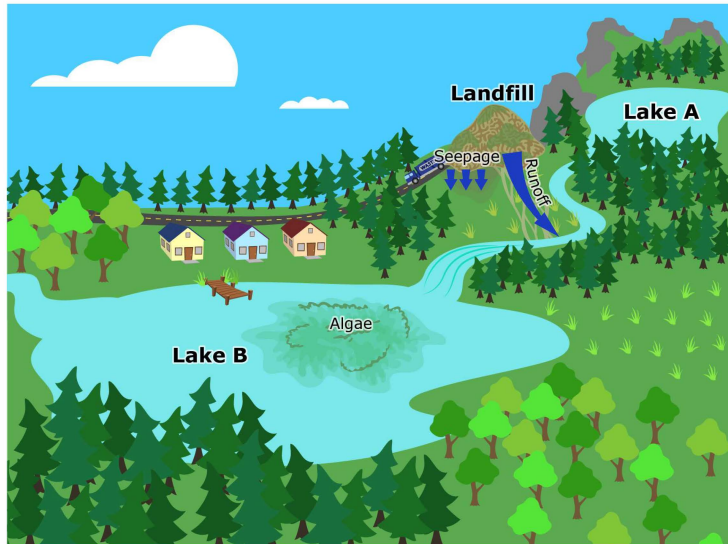
SEP2: Developing and using models

CCC4: Systems and system models

Achievement Level: At State Expectations

Algae Bloom

Clair and Rashaun want to swim in Lake B near their homes, but it is covered with algae. However, Lake A, a few kilometers away, has clear water. Clair learns that pollutants from landfills can seep through the soil to reach the groundwater and also enter nearby streams as runoff.



Clair and Rashaun hope to find a solution to the algae problem. They listed their solutions in a table.

Solution	Image
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Plastic Liner	
Shallow Channel	

3. Clair and Rashaun decide to model how the pollutants make their way from the landfill to the lake.

What is the correct path that the pollutants take? Enter 1, 2, 3, or 4 into each box to show the order of steps in the path, with 1 being the first step and 4 the last.

Lake B	4	1 point for all four in the correct order
Runoff	2	
Stream	3	
Waste from human activities	1	

Standards Alignment

Discipline: Earth and Space Science
NGSS Topic: Earth's Systems

DCI: ESS3.C

Human activities in agriculture, industry, and everyday life have had major effects on the land, vegetation, streams, ocean, air, and even outer space. But individuals and communities are doing things to help protect Earth's resources and environments.

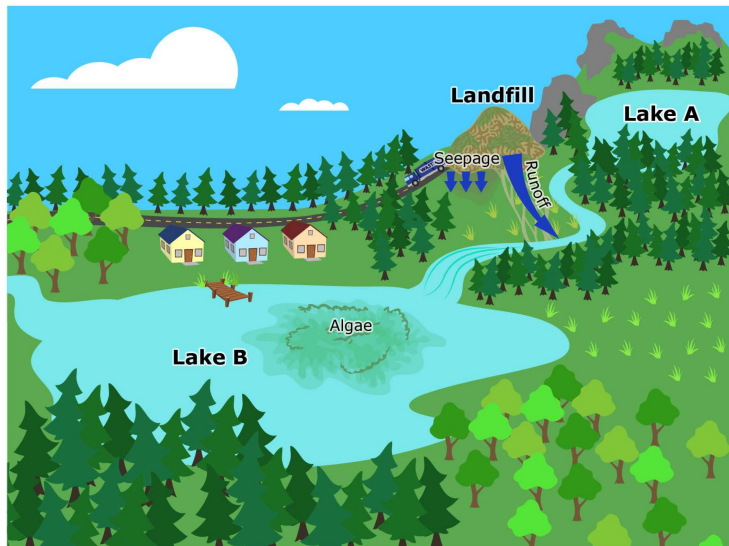
SEP2: Developing and using models

CCC4: Systems and system models

Achievement Level: Below State Expecations

Algae Bloom

Clair and Rashaun want to swim in Lake B near their homes, but it is covered with algae. However, Lake A, a few kilometers away, has clear water. Clair learns that pollutants from landfills can seep through the soil to reach the groundwater and also enter nearby streams as runoff.



Clair and Rashaun hope to find a solution to the algae problem. They listed their solutions in a table.

Solution	Image
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4. After considering the different solutions to the problem, Clair decides that using a raised barrier will best prevent pollutants from the landfill from reaching Lake B.

Which statement best states whether this solution will work?

- A** The solution will work because a raised barrier reduces runoff.
- B** The solution will work because a raised barrier reduces seepage.
- C** The solution will not work because a raised barrier will not reduce runoff.
- D** The solution will not work because a raised barrier will not reduce seepage.

1 point

Standards Alignment

Discipline: Earth and Space Science
NGSS Topic: Earth's Systems

DCI: ESS3.C

Human activities in agriculture, industry, and everyday life have had major effects on the land, vegetation, streams, ocean, air, and even outer space. But individuals and communities are doing things to help protect Earth's resources and environments.

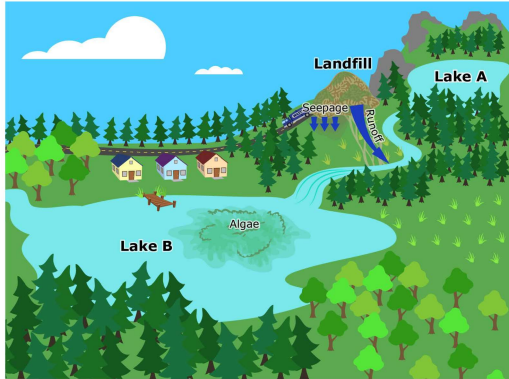
SEP2: Developing and using models

CCC6: Structure and Function

Achievement Level: Above State Expectations

Algae Bloom

Clair and Rashaun want to swim in Lake B near their homes, but it is covered with algae. However, Lake A, a few kilometers away, has clear water. Clair learns that pollutants from landfills can seep through the soil to reach the groundwater and also enter nearby streams as runoff.



Clair and Rashaun hope to find a solution to the algae problem. They listed their solutions in a table.

Solution	Image
Raised Barrier	A cross-section diagram showing a hill of 'Waste' on the right. On the left, a brick 'Raised Barrier' is built up to prevent waste from reaching the ground level.
Plastic Liner	A cross-section diagram showing a hill of 'Waste' on the right. On the left, a 'Plastic Liner' is shown as a layer of plastic material placed between the waste and the ground.
Shallow Channel	A cross-section diagram showing a hill of 'Waste' on the right. On the left, a 'Shallow channel' is shown as a ditch dug into the ground to divert runoff away from the lake.

5. Clair and Rashaun researched different solutions to prevent the pollution in the landfill from reaching Lake B.

Which solution from the table is the best solution to prevent algae from growing in Lake B? Identify a method and explain how this would prevent algae growth in Lake B.

B *I* U | ☰ ☰

2 points possible, can earn partial credit

See next page for rubric.

Standards Alignment

Discipline: Earth and Space Science

NGSS Topic: Earth's Systems

DCI: ESS3.C

Human activities in agriculture, industry, and everyday life have had major effects on the land, vegetation, streams, ocean, air, and even outer space. But individuals and communities are doing things to help protect Earth's resources and environments.

SEP2: Developing and using models

CCC6: Structure and function

Achievement Level:

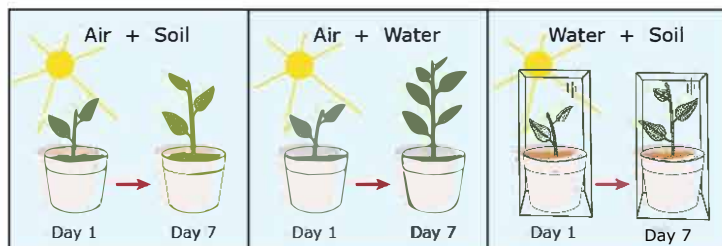
1 point = Below State Expectations

2 points = At State Expectations

Points	Qualities of the Student Response
2	The response must identify the plastic liner as the best solution to prevent algae growth and explain that the plastic liner is an impermeable barrier that will help prevent pollutants from entering the groundwater that could feed into the creek or lake B. <u>Example Student Response:</u> The plastic liner because the pollutants cannot penetrate it and enter the water around lake B.
1	The response demonstrates a partial understanding of the prompt. The response must include only one of the following: • identifies the plastic liner but does not provide an explanation. • provides an explanation but does not specifically identify the plastic liner
0	The response demonstrates minimal understanding of the prompt. The response is incorrect or contains some correct work that is irrelevant to the skill or concept being measured.

Keaton's family has many plants inside their house and outside in their garden. Keaton sees that some plants grow faster and look healthier than others. This causes Keaton to wonder which factors affect plant growth. He decides to perform an investigation. He places three similar plants in similar pots on the same windowsill. However, he changes some of the conditions under which the plants will grow for the next seven days.

Keaton makes a poster showing the results after seven days.



6. Keaton uses the data from his investigation to identify the factors that help plants grow.

Which two factors help plants grow the most? Choose two.

☒ A	air
☐ B	soil
☐ C	sunlight
☒ D	water

1 point for both correct factors

Standards Alignment

Discipline: Life Science

NGSS Topic: Matter and Energy in Organisms and Ecosystems

DCI: LS1.C

Plants acquire their material for growth chiefly from air and water.

SEP4: Analyzing and interpreting data

No CCC

Achievement Level: Below State Expectations